

PumpGuard Ops

24-pump predictive-maintenance pilot

Earlier review of emerging pump risk

Board proposal for a controlled 12-month pilot

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Decision requested

A capped pilot with two release gates

KES 1.50M

maximum first-year investment

Approve the 24-pump pilot subject to a data audit and 90-day shadow validation.

Gate 1

Sensor and maintenance data pass completeness and lineage checks.

Gate 2

Recall reaches 80% and false-positive rate remains below 5% on real pilot data.

Scale only after verified savings exceed the annual operating cost.

Maintenance triage problem

The financial exposure comes from delayed action, not spreadsheet labour

24

pumps in the proposed pilot

30 min

assumed daily manual review

8

qualifying failure events per year assumed

KES 1.2M

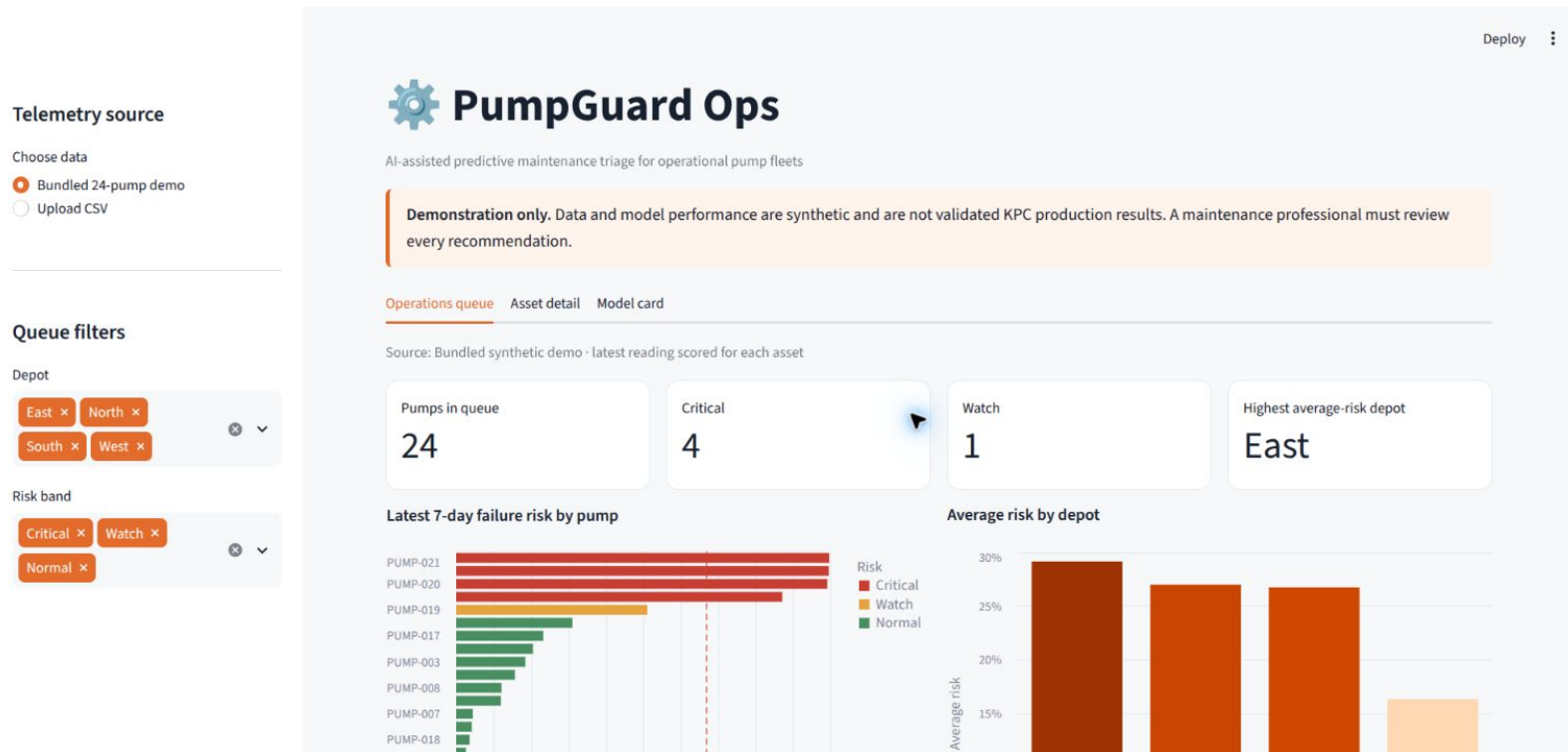
average avoidable impact per event assumed

The maintenance team must compare several sensor streams before deciding which asset needs attention. Delayed investigation can turn an emerging fault into lost production, emergency work, or secondary damage.

All figures on this slide are pilot planning assumptions. The data audit must replace them with verified operating history before Gate 1 approval.

PumpGuard operating workflow

Telemetry becomes an engineer-reviewed action queue



1. Validate

Reject missing fields, invalid values, and duplicate readings.

2. Prioritize

Score the latest reading and rank pumps by seven-day risk.

3. Review

Show recent telemetry and the signals contributing to model risk.

Engineers retain every maintenance and shutdown decision.

Prototype evidence

Synthetic grouped evaluation tests the workflow, not production readiness

0.943

ROC-AUC

81.5%

recall

2.7%

false-positive rate

13

automated tests

The model learned from 2,496 historical readings across 24 synthetic pumps. Grouped validation kept each pump in one fold, which reduces leakage from repeated time-series observations.

Production proof still requires real failure labels, probability calibration, drift monitoring, and engineering review.

Base-case financial return

Annual benefit of KES 2.575M against KES 1.500M first-year cost

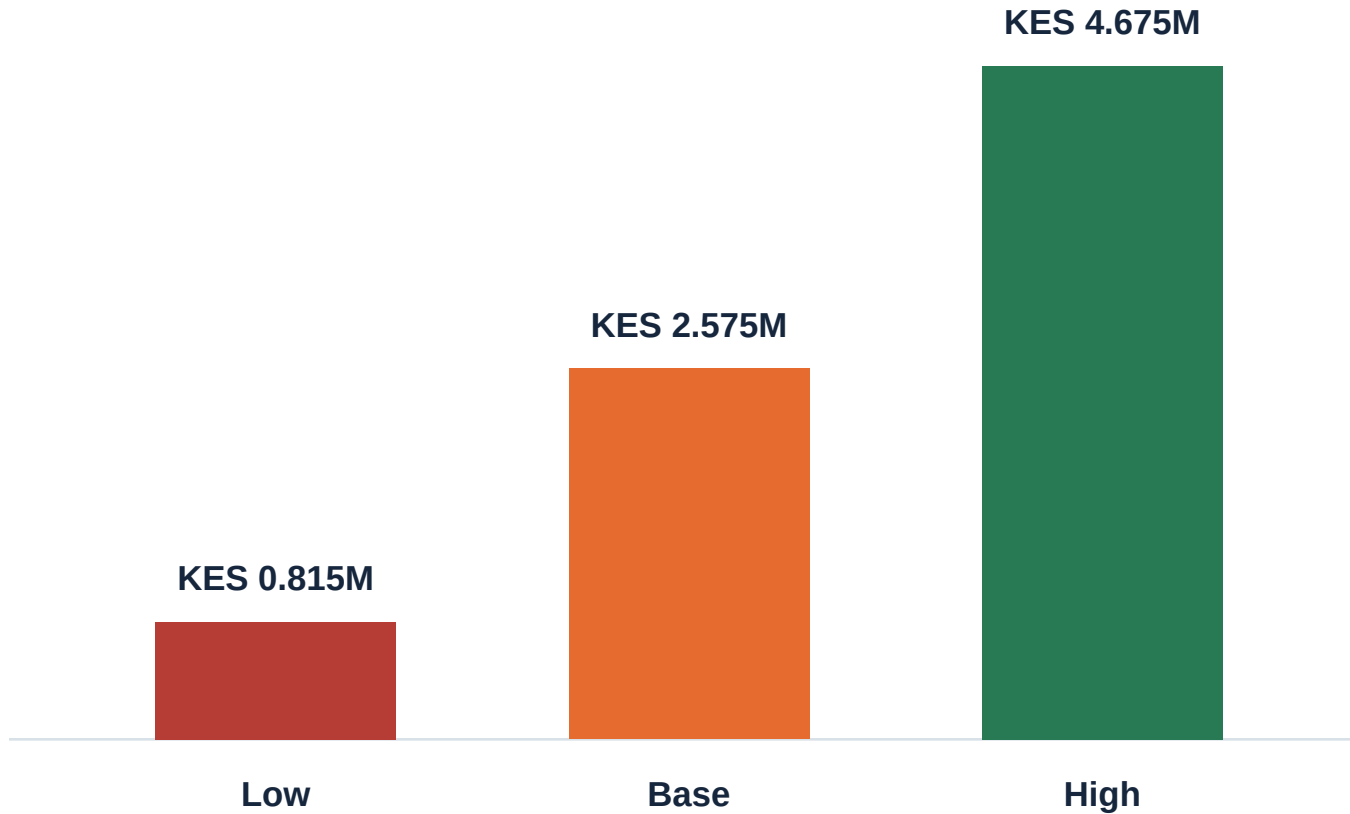
Calculation	Formula	KES
Avoided disruption	8 events x 25% x 1.20M	2.400M
Review labour saved	28 min x 250 days x 1,500/hr	0.175M
Gross annual benefit	2.400M + 0.175M	2.575M
First-year cost	0.900M setup + 0.600M run	1.500M
Net first-year benefit	2.575M - 1.500M	1.075M

7.0 mo
payback

72%
first-year ROI

Return sensitivity

The pilot breaks even only when operational losses fall



Low: 4 events x 20% x KES 0.80M + labour saving

Base: 8 events x 25% x KES 1.20M + labour saving

High: 10 events x 30% x KES 1.50M + labour saving

Low case ROI: -46%

Base case ROI: 72%

High case ROI: 212%

Pilot risks and controls

Risk	Financial or operational effect	Control and decision evidence
Model underperforms on real pumps	Missed faults or unnecessary inspections	90-day shadow run; engineer labels; 80% recall and 5% false-positive gates
Sensor data is incomplete	Unreliable ranking and wasted integration spend	Data audit before release; completeness dashboard; reject structurally invalid records
Team over-relies on risk scores	Unsafe automated action or weak accountability	Advisory queue only; named engineering approver; retain inspection evidence
Benefits cannot be verified	Pilot continues without an economic return	Log avoided events, downtime, review hours, and interventions against a baseline

Twelve-month pilot plan

Funding and operational authority increase only after evidence



Months 1-2 Data readiness

Confirm sensor coverage, maintenance labels, security, and baseline costs.



Months 3-4 Shadow validation

Run without changing maintenance decisions. Measure recall and false alarms.



Months 5-6 Controlled field use

Prioritize one maintenance team. Require engineer approval for every action.



Months 7-12 Value verification

Compare avoided events, downtime, review time, and operating cost with baseline.

Board recommendation

Approve the pilot, retain the gates, and require measured value

Approve up to KES 1.50M

for a 24-pump, 12-month controlled pilot

Data readiness before integration funding

Real-data performance before operational use

Verified net savings before scale

Base case

72% ROI

7.0-month payback